

System Diagnostics & Platform Hardening

AI Idea Forge — QA Resolution Report (Batch 4 – Group 8)

[STATUS: STABLE]

[TOTAL ANOMALIES
RESOLVED: 19]

[TIMESTAMP:
August 1, 2026]

QA Telemetry: 19 Vulnerabilities Resolved

Domain 1: Structural Integrity (Backend & State)



Resolved duplicates, poll loops,
webhook retries

Domain 2: Perimeter Defense (Validation & Auth)

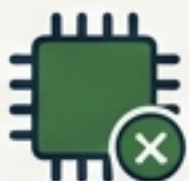


Resolved URL parsing, unauthorized
access, raw error leaks

Domain 3: Application Flow (UI/UX Logic)



Resolved state restoration,
recommendation logic, empty forms



0 Memory Leaks Remaining
System Integrity: Verified



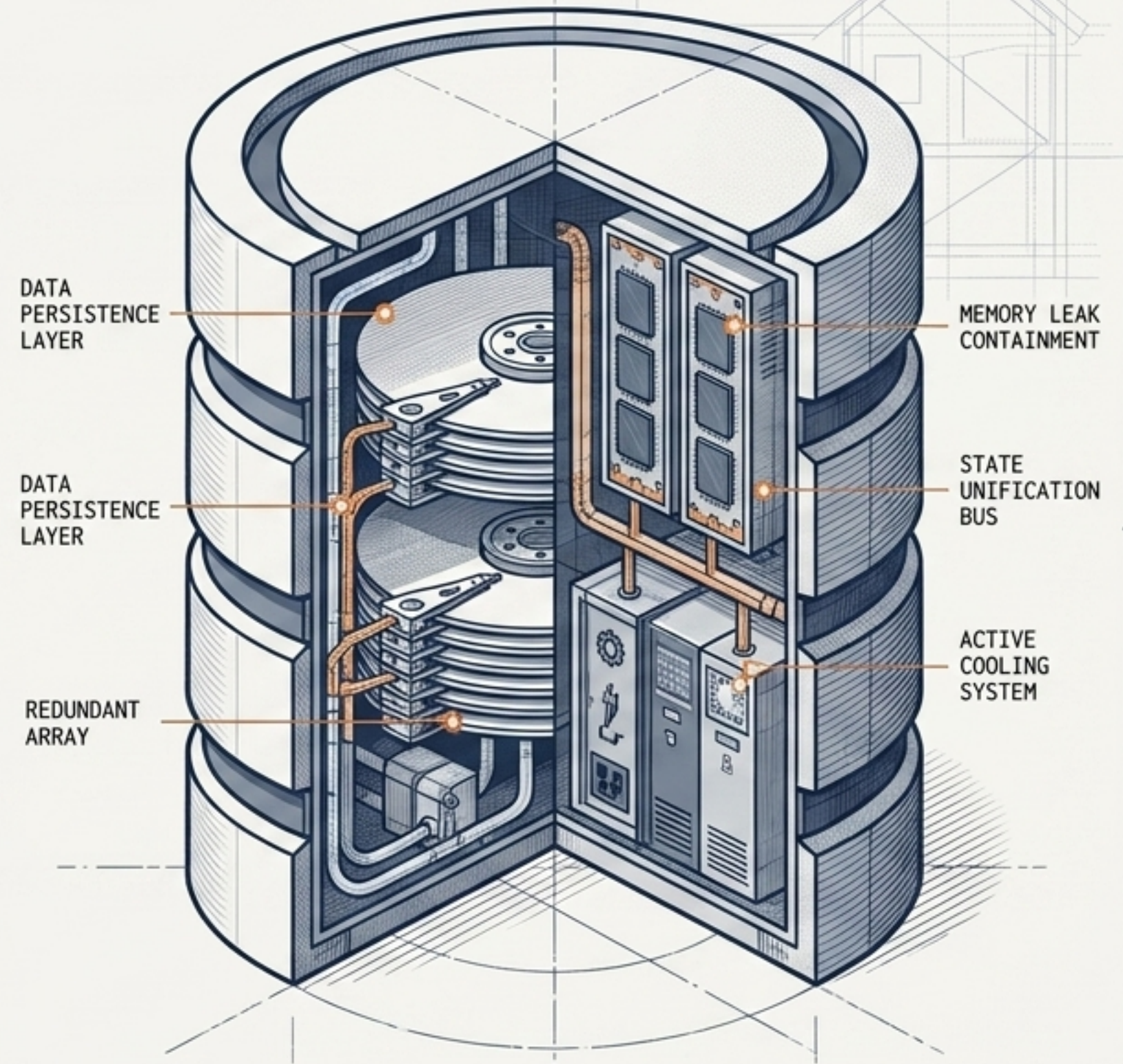
100% Protected Routes
Access Control: Enforced



Strict Type Validation Active
Data Consistency: Assured

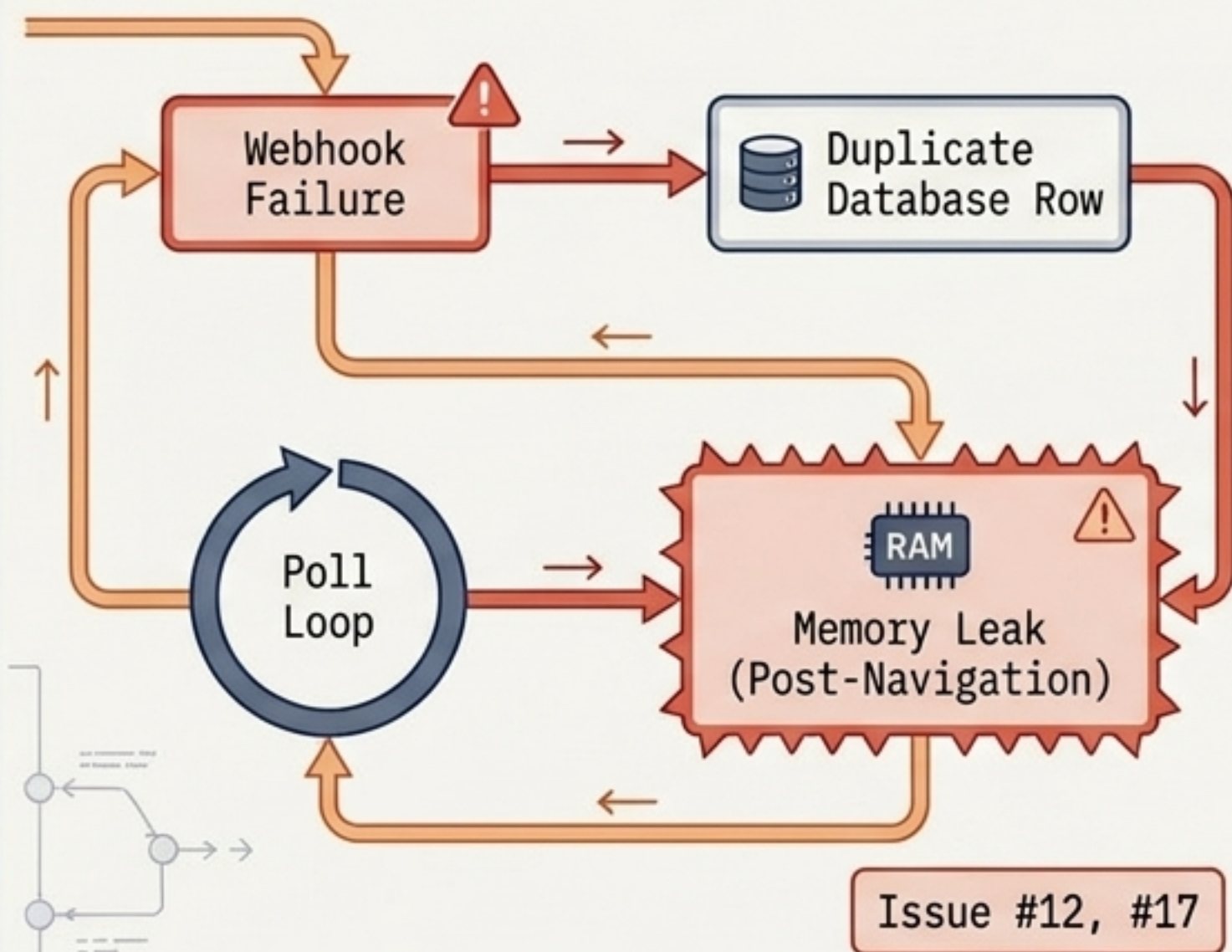
I. Structural Integrity

Fortifying data persistence, eliminating memory leaks, and unifying application state.

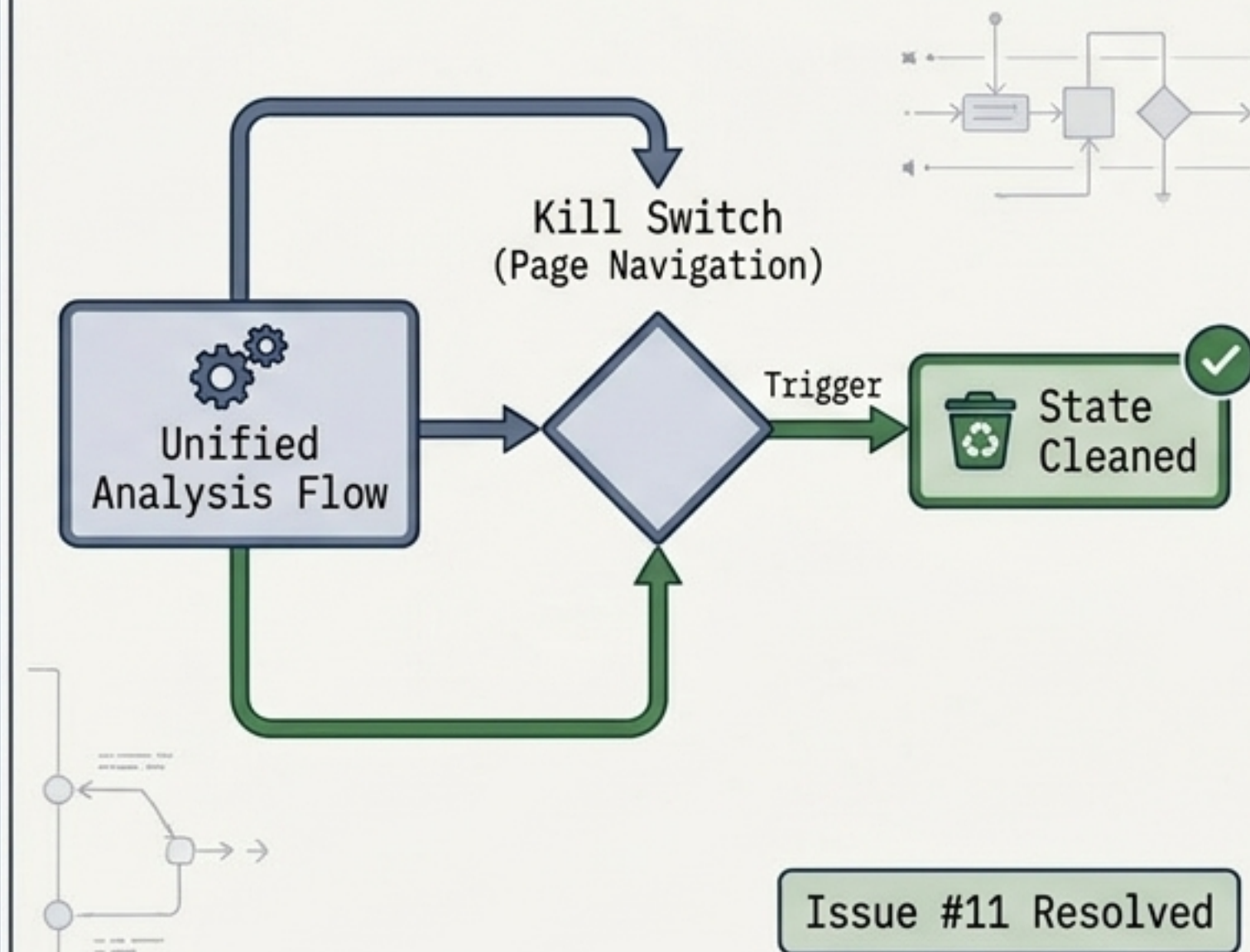


Eradicating Memory Leaks and Disconnected Records

Panel A: The Leaky Loop (Before)



Panel B: The Idempotent Lifecycle (After)



Database State Management



Diagnostic Table

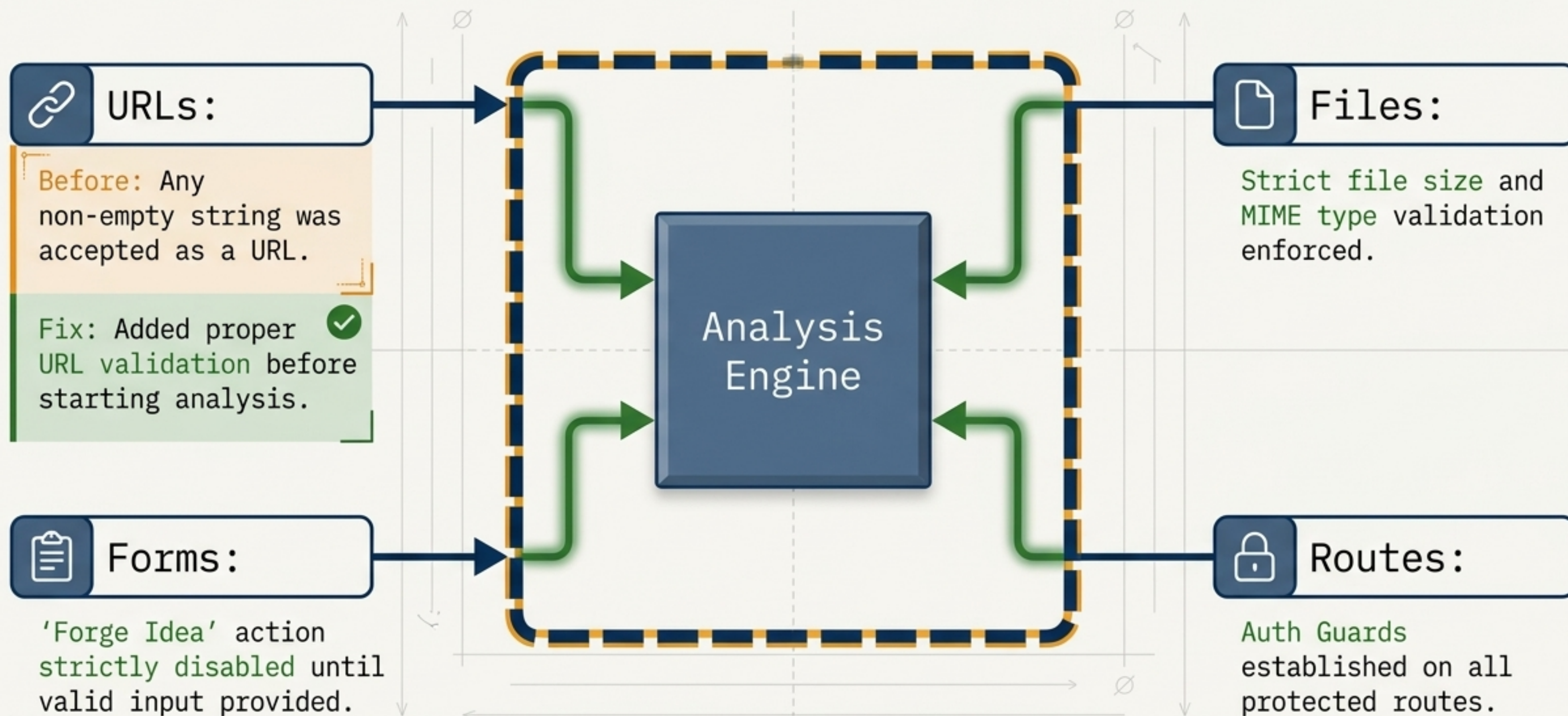
Symptom (Observation)	Remedy (Architecture)
Duplicate records were created for the same analysis.	Reused the existing analysis_id instead of creating duplicate records using application logic.

II. Perimeter Defense

Sealing input vulnerabilities, establishing strict validation, and masking internal database architectures from end users.



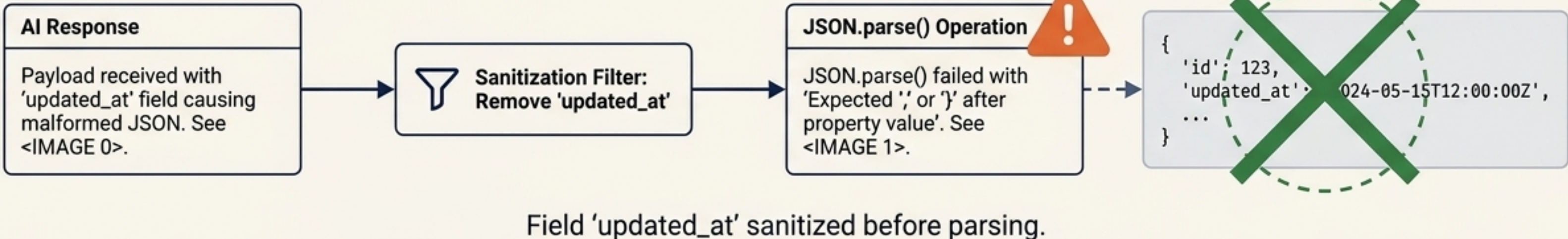
The Input Firewall Matrix



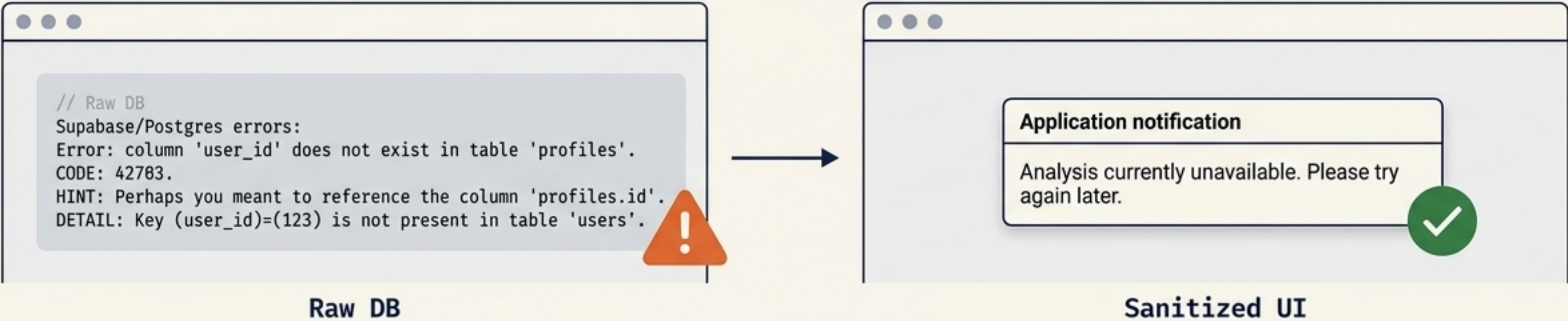
Payload Sanitization & Security



The n8n JSON Pipeline

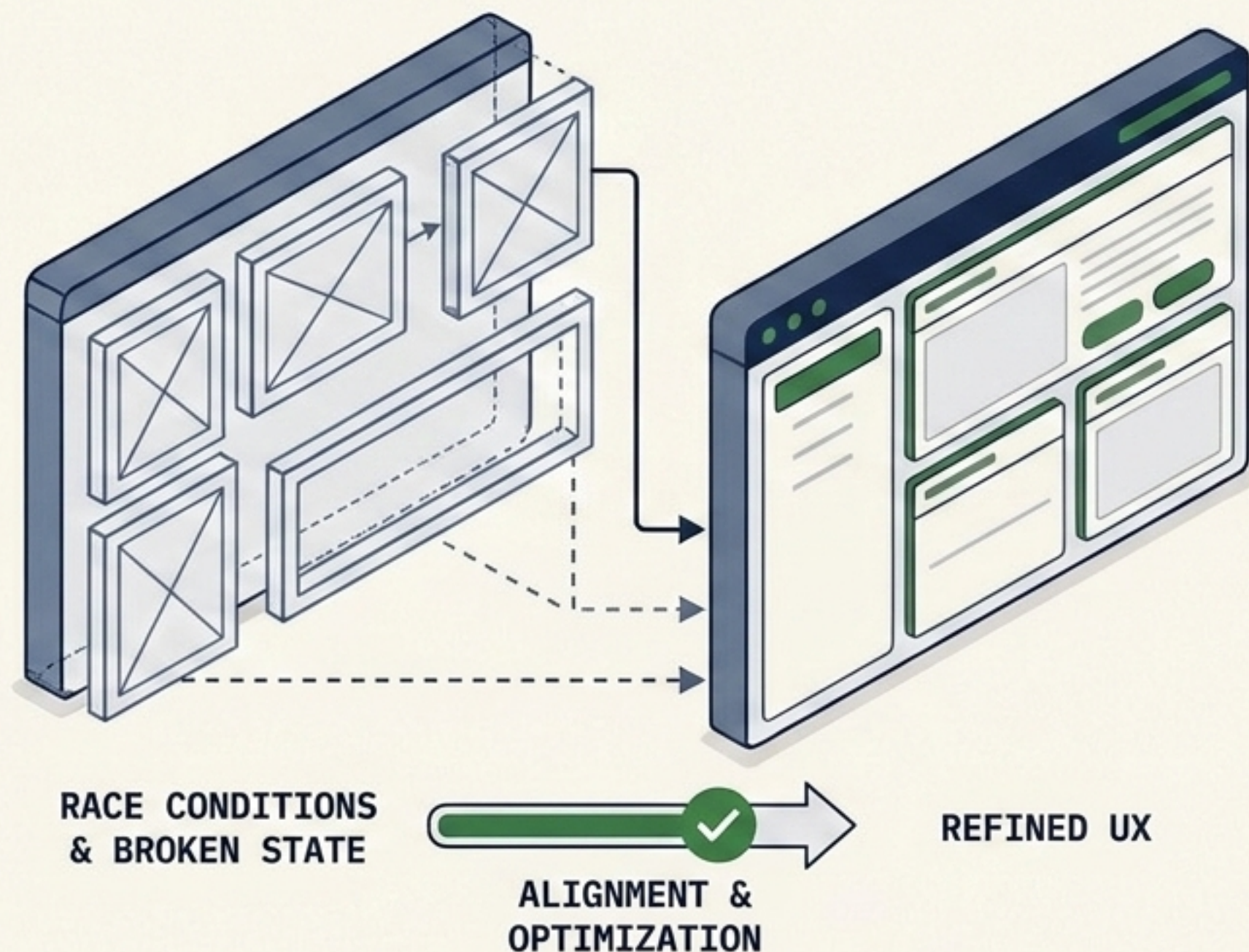


Error Masking

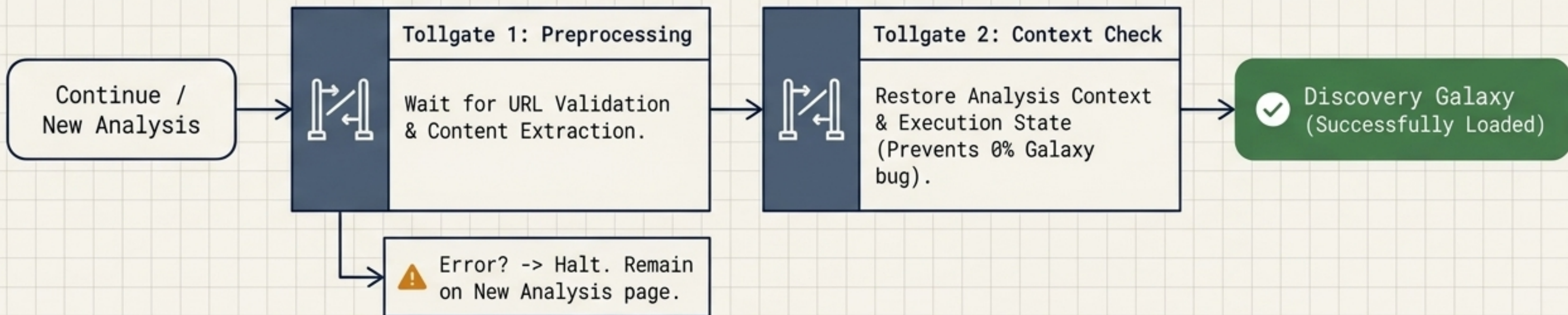


III. Application Flow & Interface Logic

Resolving race conditions in navigation, aligning visual hierarchies, and refining edge-case UX.



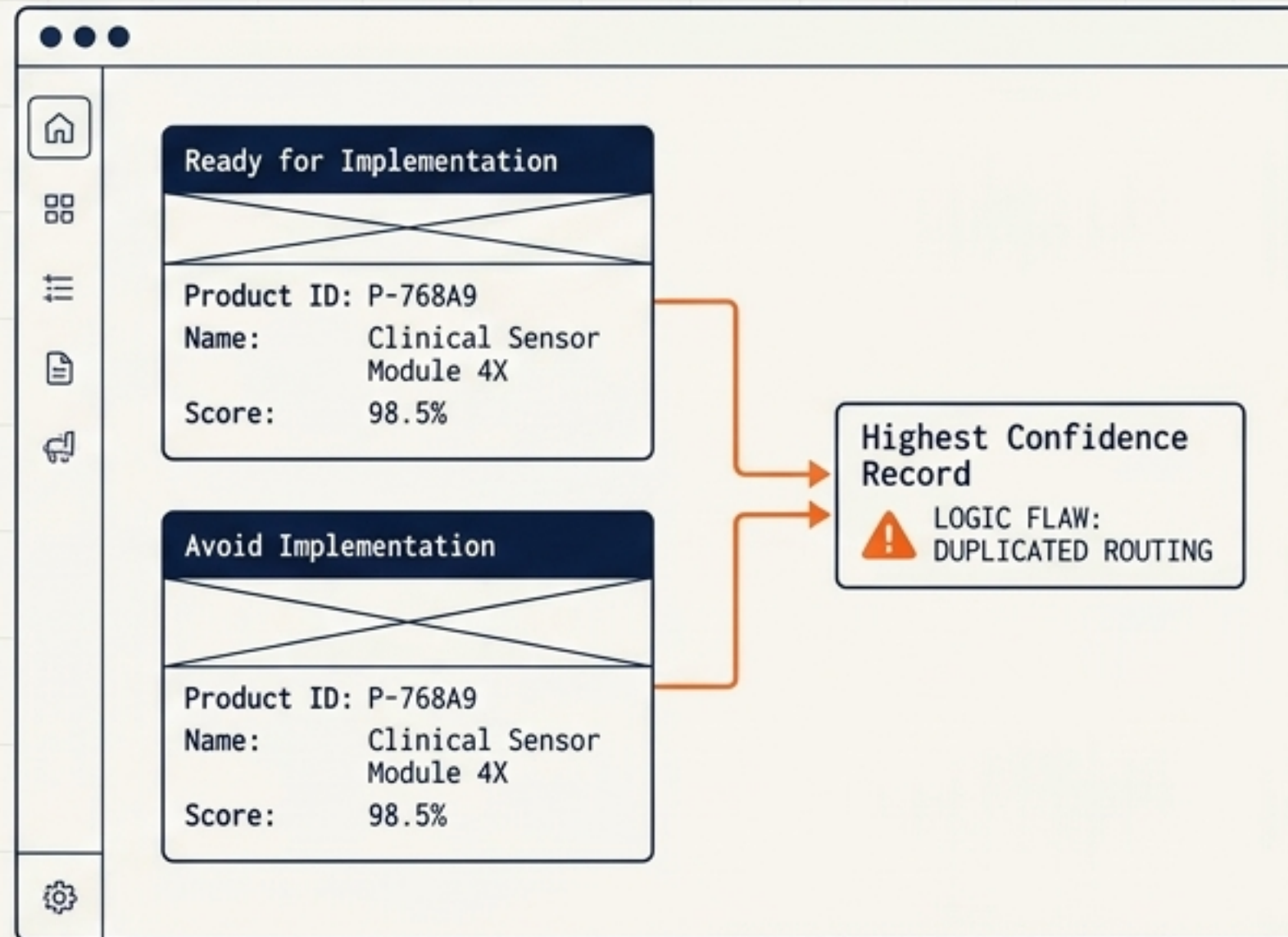
Navigational Tollgates & State Restoration



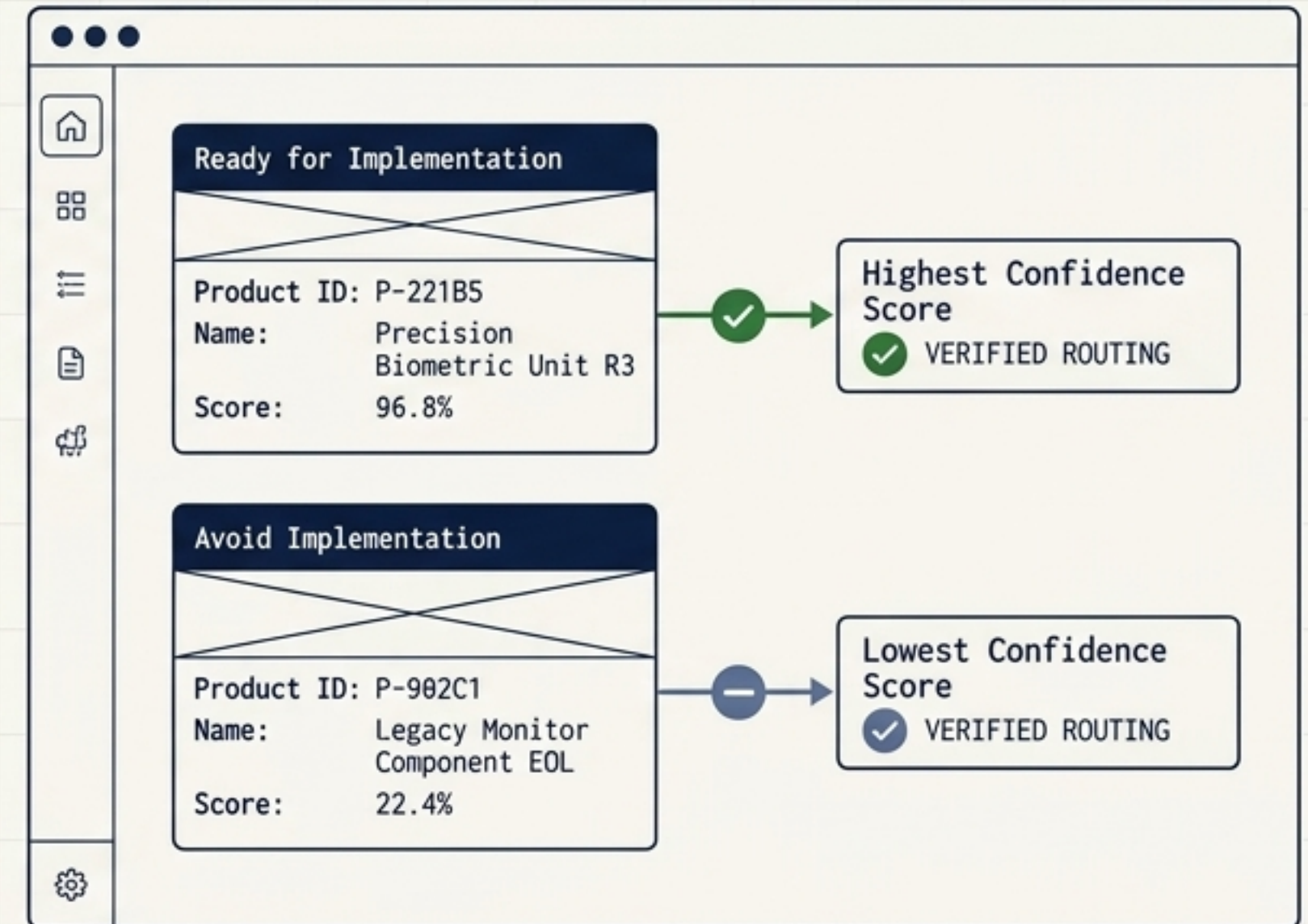
Confidence Scoring Logic

Issue #3 Resolved

BEFORE



AFTER



Interface Refinement & Edge Cases

Redundancy

Removed overlapping 'Add Notes' option to streamline the 'Your Idea' tab.



Silent Failures

Added explicit error messages when report generation downloads fail.



Agent Resilience

Added retry actions for failed agents.



Dynamic Guidance

Enforced consistent character limits and added source-specific live validation messages.



System Stable. Ready for Deployment.



END OF REPORT. // BATCH 4 – GROUP 8